DESIGN THINKING

The design thinking (DT) process has produced countless innovations (i.e., wireless headphones, virtual reality, artificial intelligence, video streaming sites, etc.). DT offers challenging ideas used in decision-making and team management.

Design thinking is iterative. It means a person tests an assumption, returns to the prototype stage, and modifies it based on the results and feedback.

Design thinking is a state of mind. It's a human-centric, holistic approach to problem-solving and business thinking that employs empathy, ideation, prototyping, and experimentation to solve real-world issues. Design thinking works horizontally across an organization to improve communications and deliver new insights. It is called "the search for a magical balance between business and art; structure and chaos; intuition and logic; concept and execution; playfulness and formality; and control and empowerment." It is called "the glue between disciplines."

Advantages of Design Thinking:

- It assists business people in keeping their focus on humans and their needs.
- It relies on both creativity and logic.
- It promotes a learn-by-doing approach and suggests that failure is an excellent way to learn.
- It is collaborative.

Phase I: Understanding and Empathy

Empathy focuses on the human-centered design process. Empathy requires understanding how people do things, their physical and emotional needs, their thinking about the world, and what is meaningful to them.

Benefits of Empathy

Observing what people do and how they interact with their environment gives a design thinker clues about how people think and feel. Through observation, design thinkers capture physical manifestations of people's experiences (what they do and say). It translates the meaning of the experiences into insights. These insights give direction to create innovative solutions. The best solutions come out of the best insights into human behavior. Why? Because people's minds automatically filter information without them realizing it. Design thinkers need to see things "with a fresh set of eyes," and empathizing gives them those new eyes.

Engaging with people directly reveals the way they think and the values they hold. Sometimes, these thoughts and values are not obvious to the people who hold them, and a good conversation can surprise both the designer and the subject with surprising insights. The stories people tell, and the things they say (even if they differ from what they do) are strong indicators of their beliefs about the world. Good designs are built on a solid understanding of these beliefs and values.

How to Empathize

To empathize, a design thinker must:

- **Observe.** View subjects and their behavior in the context of their lives. As much as possible, do observations in relevant contexts alongside interviews. Some of the most powerful realizations come from noticing a disconnect between what someone says and what s/he does.
- **Engage** (sometimes called "interview"). Prepare questions for the subject, but expect to let the conversation deviate from them. Keep the conversation loosely bounded. Elicit stories from the

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people and always ask "why?" to uncover more profound meaning. Engagement can come through short "intercept" encounters and extended scheduled conversations.

- **Watch and listen.** Ask the subject to demonstrate how they complete a task. Have them go through the steps and explain why they do what they do. Ask them to vocalize what's going through their mind as they perform a task or interact with an object. Have a conversation in the context of someone's home or workplace—so many stories are embodied in artifacts. Use the environment to prompt deeper questions.

Phase II: Defining the Problem

The defined mode of the design process brings clarity and focus to the design space. The design thinker's responsibility is to define their challenges based on what they learned about their subjects.

Benefits of Problem Definition

This phase is critical because it results from a point-of-view (POV), the explicit expression of the problem. The POV defines the right challenge/s to address. Crafting a narrowed-down problem statement yields greater quantity and higher quality solutions when generating ideas. The defined mode is an endeavor to synthesize scattered findings into powerful insights.

How to Define

Consider what stood out to you when talking and observing people. What patterns have emerged? If you noticed something interesting, ask yourself (and your team) why that might be. You connect that person to a larger context by asking why someone had a particular behavior or feeling.

Develop an understanding of your subject (i.e., your USER or the target market). Synthesize and select a limited set of NEEDS that you think are important to fulfill; you may express just one (1) salient need to address. Work to express INSIGHTS you have developed through synthesizing information with the help of empathy and research work. Then, establish a point-of-view by combining the three (3) elements (user, need, and insight).

Phase III: Brainstorming Solutions

Ideate concentrates on idea generation. Mentally, it represents a process of "going wide" regarding concepts and outcomes. Ideation provides both the fuel and the source material for building prototypes and getting innovative solutions into the hands of the users.

Benefits of Brainstorming

Ideation combines the problem space context and the users or the target market to generate solutions to identified challenges or needs. Particularly early in a design project, ideation is about pushing for the widest possible range of ideas you can select, not simply finding the best solution. The best solution will be determined later through user testing and feedback.

The various forms of ideation are leveraged to:

- Stepping beyond obvious solutions, thus increasing the innovation potential of the solution set.
- Harnessing the collective perspectives and strengths of the team.
- Uncovering unexpected areas of exploration.
- Creating fluency (volume) and flexibility (variety) in the innovation options.
- Generating obvious solutions and implementing them.

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How to Brainstorm

Ideation starts by combining the conscious and unconscious mind with imagination. For example, during brainstorming, the synergy of the group is maximized by building on others' ideas to reach the best solutions. Adding constraints, being surrounded by related materials that inspire the team, and embracing misunderstandings allow the best results.

Another ideation technique is prototyping. In creating a tangible output, a person comes to a point where decisions must be made. It encourages new ideas to come forward.

There are other ideation techniques, such as mind mapping and sketching, where the common theme is *deferring judgment* (separating idea generation from evaluation). By doing so, imagination and creativity are given a voice.

Phase IV: Prototyping Solutions

Prototyping involves the iterative generation of artifacts and the creation of tangible output. In the early stages of a project, the design thinker should create low-resolution prototypes that are quick and cheap to make but can elicit valuable feedback from users and colleagues. In later stages, the prototype will get a little more refined as useful feedback is applied to make modifications.

Benefits of Prototype

- **To ideate and problem-solve.** Building a prototype helps design thinkers identify possible issues with the proposed product design or service model.
- **To communicate.** Interacting with users is often richer when centered around a conversation piece. A prototype is an opportunity to elicit valuable feedback from the users.
- **To fail quickly and cheaply.** Committing as few resources as possible means less time and money invested upfront.
- **To test possibilities.** Staying on low resolution allows design thinkers to pursue different ideas without committing to a direction too early on.
- **To manage the solution-building process.** Identifying a variable also encourages design thinkers to break a large problem into smaller, testable chunks.

How to Prototype

- **Start building.** Even if you are unsure what you are doing, picking up some materials—post-its, tape, and other small objects is an excellent way to start. It will be enough to get you going.
- **Don't spend too long on one (1) prototype.** Let go before you get too emotionally attached to one (1) prototype.
- **Identify a variable.** Identify what's being tested with each prototype. A prototype should answer a particular question when tested. For instance, the prototype must consider functionality, efficiency, effectiveness, etc.
- **Build with the user in mind.** What do you hope to test with the user? What sorts of behavior do you expect? Answering these questions will help focus your prototyping and help you receive meaningful feedback in the testing phase.

Phase V: Testing the Solution

The test mode includes soliciting feedback from the prototypes given to the users. It is another opportunity to gain empathy from the target market.

Testing is another opportunity to understand the users. Focus on interacting with users, but don't reduce the "testing" to asking whether or not people like the solution (product or service model). Instead,

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continue to ask "why?" and focus on what you can learn about the person, the problem, and your potential solutions.

Ideally, you can test within the real context of the user's life. For a physical object, ask people to take it with them and use it within their normal routines. For a service (experience), try to create a scenario in a location that would capture the real situation. If testing a prototype is impossible, frame a more realistic situation by having the users take on a role or task when approaching your prototype. A rule of thumb: always prototype as if you're right, but test as if you're wrong; testing is the chance to refine and improve your solutions.

Benefits of Testing the Solution

- **To refine prototypes and solutions.** Testing informs the subsequent iterations of prototypes. Sometimes, this means going back to the drawing board.
- **To learn more about your user.** Testing is another opportunity to build empathy through observation and engagement; it often yields unexpected insights.
- **To refine your POV.** Testing sometimes reveals that you got the solution wrong and failed to frame the problem correctly.

How to Test

- **Show; don't tell.** Put your prototype in the user's hands (or your user within an experience). And don't explain everything (yet). Let your tester interpret the prototype. Watch how they use (and misuse!) what you have given them and how they handle and interact with it; then, listen to what they say about it and their questions.
- **Create experiences.** Create your prototypes and test them in a way that feels like an experience your user is reacting to rather than an explanation your user is evaluating.
- **Ask users to compare.** Bringing multiple prototypes to the field for testing gives users a basis for comparison, and comparisons often reveal latent needs.

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